PROJECT DOSSIER · VERSION 1.0 · OCTOBER 2026

NeoStay

Aims, methods, and the road to a hospital-grade service

What the platform is for, the statistical and machine-learning work that will make its forecasts trustworthy, how each role uses it day to day, and everything required to install it across a hospital group with several branches.

Waseem Altaf, MD

Project and Clinical Lead · Neonatologist

Clinical direction, outcome interpretation, data stewardship, domain validation, hospital coordination.

Abdul Razak, PhD

Technical Lead, Lead Developer and Applied AI/Data Science Lead

Architecture, analytical and algorithmic design, implementation, software validation, deployment.

DOCUMENT
Detailed project dossier

AUDIENCE
Clinical leadership, nursing management, hospital IT, data partners

COMPANION DOCUMENTS
Executive brief · Data & feature requirements · Architecture reference

STATUS
Decision-support prototype — not a medical device, not clinically validated

00 Contents

Sections 1–4 describe what exists. Sections 5–6 describe what is proposed and how it will be proven.

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How to read this dossier. Statements about the **current** system were checked against the source code and are reproducible by running it. Statements about **models**, **features** and **rollout** are proposals for review: they describe what will be built, tested and reported, and each one names the evidence that would justify adopting it.

4
PREDICTION PROBLEMS

12
CANDIDATE MODEL FAMILIES

6
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3
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01 Purpose, goals and success measures

Why the project exists, and how we will know it worked.

Neonatal units plan under two kinds of uncertainty: how long each baby will stay, and how much skilled nursing the unit will need shift by shift. Both are usually answered from memory and experience. NeoStay's purpose is to answer them from the unit's own recorded outcomes and its own acuity policy, in a form a clinician can read in under a minute and a manager can defend in a staffing review.

Aims

TABLE 1 — AIMS AND THE MEASURABLE GOALS BENEATH THEM

Aim	Goal	How success is measured
Anticipate the journey	Give every admission an expected length of stay, a realistic range, and the likely discharge outcome, with an honest confidence signal.	Prediction error and calibration against the unit's own subsequent outcomes, reported per birth-weight band (Section 7).
Make staffing defensible	Convert the unit's acuity policy into nurses per shift, the required skill mix, and a roster that names gaps before the shift starts.	Agreement between NeoStay's recommended establishment and the staffing a senior charge nurse would set, measured on recorded shifts.
Protect skill mix, not just headcount	Surface when a shift is numerically adequate but too junior for the babies present.	Number of shifts flagged, and whether the flag matched the unit's own retrospective judgement.
Be usable in a busy unit	A full census classified and a roster produced in under five minutes, on a ward computer or a phone.	Timed task completion with real charge nurses; fewer than two support questions per user in month one.
Travel between hospitals	A second site adopts NeoStay by configuration — its own grades, ratios and acuity policy — not by code changes.	Time to onboard branch two, and the number of code changes required to do it (target: zero).
Be safe and auditable	Every number traceable to a source; access restricted to named accounts; no patient-level data held.	Successful information-governance review at the host hospital, and a clean security review.

Who benefits, and how

The baby and family

Earlier, better-informed conversations about likely length of stay, and a unit more reliably staffed to the acuity actually present.

The nursing team

A transparent basis for the shift's establishment and skill mix, and an explicit record of what was requested when cover was short.

The hospital

Objective evidence for establishment reviews, bank and agency decisions, and cross-branch comparison of demand.

A deliberate limit. NeoStay recommends; it never assigns. Final decisions on cot allocation, escalation and individual nurse assignment rest with the charge nurse and the consultant on duty. The interface is written so that this is never ambiguous.

02 Scope, and what this project is not

Boundaries agreed before any data arrives.

In scope

- Forecasting length of stay, discharge outcome and survival for very low birth weight infants.
- Translating an acuity policy into nurse:patient ratios and minimum nurse competency.
- Unit-level demand, skill mix and a 24-hour roster of the nurses actually available.
- Population analytics across every birth-weight band and gestational week.
- Multi-branch deployment, per-branch configuration, and roll-up reporting.

Out of scope

- Diagnosis, treatment recommendation, drug dosing, or any individual clinical decision.
- Storing patient-identifiable records, notes or images.
- Replacing the hospital's rostering or payroll system of record.
- Automatic assignment of named nurses without human approval.
- Regulatory medical-device claims (see Section 13 for the boundary and what would cross it).

Principles that constrain every design decision

Principle	What it means in practice
Traceability before cleverness	Every figure can be traced to a source row, a policy line, or a stated assumption. A model that cannot explain a number does not ship.
Aggregate by default	The product runs on aggregated statistics. Patient-level data, where it is used for model development, stays inside the hospital's governance boundary.
Local policy wins	Ratios, grades, supervision rules and levels of care are the hospital's, not the software's.
Degrade honestly	Low sample size, missing data or an unfamiliar case is shown as such, never smoothed into false confidence.
Portable by construction	One container, environment-variable configuration, no managed cloud service required.

03 Where the project stands today

Verified against the code, October 2026.

712 VLBW INFANTS BEHIND THE STATISTICS

108 ACUITY FINDINGS ENCODED

64 LEVEL-OF-CARE CRITERIA

105 AUTOMATED TESTS PASSING

What works now

TABLE 2 — CURRENT CAPABILITY, AND THE METHOD BEHIND IT

Capability	Method today	Maturity
Length of stay, discharge outcome, survival	Linear interpolation between birth-weight and gestational-age band statistics (median, Q1, Q3) from one centre, 2016–2026; the two views are averaged; outcome shares come from recorded counts; confidence is graded by sample size.	Working, deterministic
Per-infant nursing workload	A severity score from weight and gestational age, raised by admission condition and respiratory support, splits the stay into intensive, intermediate and convalescent phases at 1:1, 1:2 and 1:3.	Working, assumption-based
Acuity classification	The hospital's acuity tool, transcribed to data: the most intensive finding across 11 body systems sets the ratio; the highest level-of-care criterion sets N1–N4; the minimum nurse level is the higher of the two.	Working, policy-driven
Unit staffing and rostering	Demand summed across cots; skill mix accumulated from the expert tier downward; novice cap and preceptor rule applied; nurses dealt across shifts and matched to cots, with uncovered cots named.	Working, rules-based
Population analytics	Length of stay, discharge outcome and survival charted across every birth-weight band and gestational week, from the same aggregated tables the forecast uses.	Working
Access control	Named accounts, PBKDF2-hashed passwords, signed expiring session cookies, rate-limited sign-in; a published copy can be encrypted so only accounts can open it.	Working
Deployment	One container: hospital server with Docker Compose and HTTPS, any Kubernetes cluster, or a serverless container host with no writable disk.	Working

The honest gap. Nothing in the current system is a fitted statistical model. Predictions are interpolations of recorded group statistics, and the staffing model rests on stated assumptions. No prospective validation has been performed. Sections 5 to 7 exist to close that gap.

04 The analytical problem, stated precisely

Four questions, four modelling tasks, four kinds of evidence.

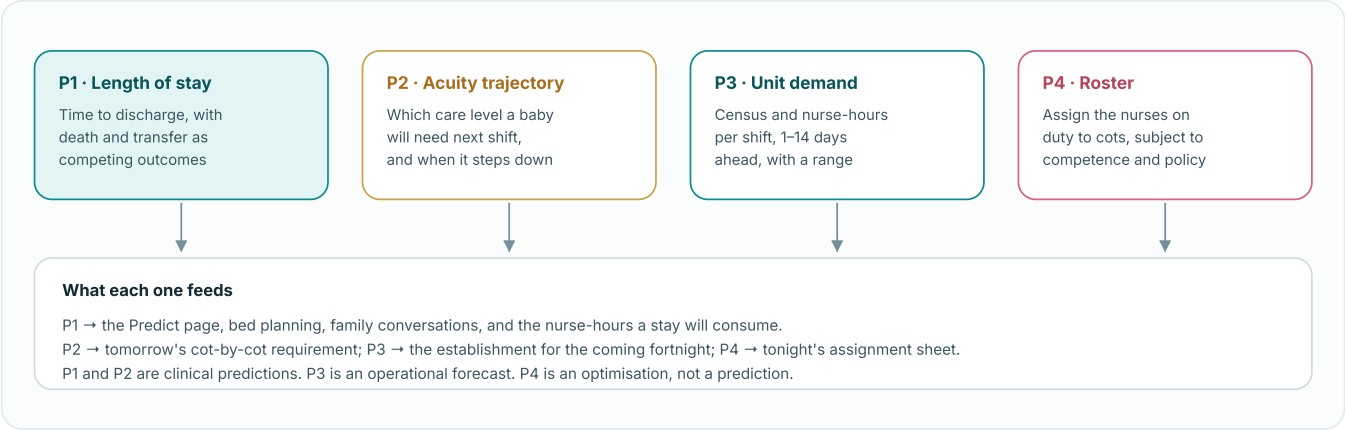


Figure 1 — The four problems. They are deliberately separated: each has its own data requirement, its own metric, and its own failure mode.

TABLE 3 — PROBLEM DEFINITIONS

Ref	Prediction target	Unit of analysis	Primary metric	Baseline to beat
P1	Days to discharge; competing outcomes home / transfer / death	One admission	Concordance and calibration; absolute error in days	Today's band interpolation
P2	Care level (N1–N4) and ratio at the next shift	One infant-shift	Balanced accuracy; cost-weighted error (under-calling is worse)	Carry forward today's level
P3	Census and total nurse-hours per shift, 1–14 days ahead	One unit-shift	Mean absolute error; pinball loss for the range; coverage	Seasonal naive (same shift last week)
P4	Feasible assignment maximising covered cots and skill fit	One shift	Uncovered cots; constraint violations; planner acceptance rate	Today's greedy rule-based assignment

05 Models to be built and tested

Shortlists, with the reason each is a candidate and the evidence that would retire it.

Method, in one paragraph. For each problem we start from the simplest model that could work, add complexity only when it earns its place on held-out data, and prefer models whose output can be explained at the cot side. Every candidate is scored on the same split, against the same baseline, with the same reporting checklist. A model that wins on error but cannot be calibrated or explained does not go into the product.

P1 · Length of stay and discharge outcome

TABLE 4 — CANDIDATE MODELS FOR P1

Model family	Why it is a candidate	Main risk
Parametric survival (AFT: log-normal, Weibull)	Length of stay is a time-to-event with censoring (still inpatient at extract). AFT gives a median and quantiles directly, which is what the interface already shows.	Distributional assumption may not hold across weight bands.
Cox proportional hazards	Standard, well-understood, no distributional assumption; good for effect sizes and for clinical review of which variables matter.	Needs a baseline hazard to produce days; proportionality often violated by respiratory support.
Competing-risks (Fine–Gray, cause-specific hazards)	Discharge home, transfer and death compete. Treating death as censoring biases stay estimates upward — the clinically important correction.	More parameters, needs enough events in each outcome.
Multi-state model	Represents the real pathway: intensive → intermediate → convalescent → discharge, with readmission to a higher level. Produces P1 and P2 from one structure.	Data hungry; needs reliable daily level-of-care records.
Gradient boosting (XGBoost / LightGBM) with survival objective	Handles non-linearity and interactions (weight × respiratory support) without hand-crafting; strong tabular baseline.	Needs calibration and explanation work; risks fitting one unit's practice.
Quantile regression / quantile forests	The product shows a range, not a point. Modelling Q1 and Q3 directly optimises exactly what is displayed.	Quantile crossing; less familiar to clinical reviewers.
Bayesian hierarchical model	Partial pooling across branches: a small unit borrows strength from the group without losing its own signal. Natural uncertainty intervals.	Slower to fit; requires care in prior choice and explanation.
Conformal prediction wrapper	Distribution-free intervals with a guaranteed coverage rate, on top of whichever model wins.	Intervals can be wide; needs a clean calibration set.

P2 · Acuity trajectory

Model family	Why it is a candidate
Markov / multi-state transition model	Care level is a small ordered state space; transition probabilities per shift are directly interpretable and easy to audit with the clinical lead.
Ordinal logistic regression	Respects the ordering N1 < N2 < N3 < N4 and gives odds ratios clinicians can challenge.
Gradient boosting with cost-sensitive loss	Under-calling acuity is more harmful than over-calling; the loss can be weighted accordingly.
Sequence model (GRU/temporal convolution)	Only if daily records prove rich enough; held as a stretch option, never as the first choice.

P3 · Unit demand forecasting

Model family	Why it is a candidate
Seasonal naive and moving average	The baseline every other model must beat; often strong for ward census.
SARIMA / state-space (ETS, structural time series)	Captures weekday and seasonal patterns with interpretable components and prediction intervals.
Gradient boosting on calendar and census features	Easily absorbs admissions pipeline, bed closures, school holidays, local events.
Queueing approximation (Erlang / M/M/c)	Converts a demand distribution into a staffing level at a stated risk of shortfall — the question management actually asks.
Discrete-event simulation	Stress-tests a proposed establishment against arrival and length-of-stay distributions; answers "what if two cots close".

Selection rule, agreed in advance. A candidate replaces the incumbent only if it improves the primary metric on a temporal hold-out *and* on at least one external branch, keeps calibration within agreed bounds, produces an explanation the clinical lead accepts, and runs inside the interaction budget (Section 12). Otherwise the simpler incumbent stays.

P4 · Roster optimisation

Approach	Why it is a candidate
Constraint programming / integer programming (CP-SAT, MILP)	Hard rules (competence, ratios, one assignment per nurse, charge nurse free) as constraints; soft goals (continuity of care, fairness of nights, preferences) as weighted objectives. Produces a provably feasible roster or a precise reason none exists.
Greedy heuristic (today's algorithm)	Instant, explainable, already in production; remains the fallback when the solver cannot finish inside the interaction budget.
Local search / simulated annealing	Improves a feasible roster within a time budget; useful for large units with many soft preferences.

06 Features: the variables the models need

What each model would learn from, grouped by where it comes from.

Feature choice is a clinical decision as much as a statistical one. The list below is the proposed candidate set, drawn from established neonatal risk scores (CRIB-II, SNAPPE-II) and from the operational variables a charge nurse already uses. The companion **Data & Feature Requirements** document turns this into a request to hospital management, field by field.

TABLE 5 — CANDIDATE FEATURE SET

Group	Features	Serves	Priority
At birth	Birth weight; gestational age (weeks + days); sex; singleton or multiple; antenatal steroids; mode of delivery; place of birth (inborn or outborn); Apgar at 1 and 5 minutes; admission temperature; birth head circumference.	P1, P2	Must
Admission severity	Base excess or pH in the first hour; FiO_2 and oxygenation index; blood pressure support; CRIB-II or SNAPPE-II where recorded; major congenital anomaly.	P1, P2	Must
Clinical course	Respiratory support type and duration by day (ventilation, HFOV, NIPPV, CPAP, high flow, nasal cannula, room air); surfactant doses; inotropes; sepsis episodes (culture-positive and clinical); NEC stage; IVH grade; PDA treatment; ROP treatment; BPD at 36 weeks; surgery; therapeutic hypothermia.	P1, P2	Must
Feeding and growth	Day of first enteral feed; day of full enteral feeds; parenteral nutrition days; central line days; weight at 7, 14, 28 days and at discharge.	P1, P2	Should
Daily acuity	Level of care per shift (N1–N4); nurse:patient ratio applied; the acuity-tool findings ticked; assessment frequency; 1:1 criteria met.	P2, P3, P4	Must
Unit operations	Hourly or shift census; admissions, discharges, transfers in and out, deaths with timestamps; cot capacity and any closures; occupancy; refused admissions.	P3	Must
Workforce	Establishment by grade; per-shift actual staffing by grade; bank and agency use; sickness and vacancy rate; preceptor pairings; contracted hours; qualified-in-speciality proportion.	P3, P4	Must
Branch context	Branch identifier; designated level of the unit; catchment; transfer agreements; local ratio policy; local grade ladder.	All, and multi-site pooling	Must
Calendar	Date, weekday, public holiday, season; school holidays; local events affecting transfers.	P3	Should
Outcome labels	Discharge date and destination; death date; transfer destination and reason; readmission within 28 days.	P1	Must

Features deliberately excluded. Ethnicity, postcode, religion, insurance status and any socio-economic proxy are excluded from the modelling set unless the hospital and its ethics process explicitly request their inclusion for an equity analysis — and then only as a fairness audit variable, never as a predictor that changes a staffing recommendation.

07 Validation, governance and monitoring of models

How a candidate earns its place, and how it is watched afterwards.

Validation protocol

01 **Split by time, not at random.** Train on the earlier period, test on the most recent 12 months, so the evaluation answers "would it have worked last year".

02 **Hold out a branch.** With more than one site, leave-one-branch-out tells us whether the model transfers or has learned one unit's habits.

03 **Report discrimination and calibration together.** Concordance or balanced accuracy alone is not enough: a calibration plot and slope decide whether numbers can be shown to clinicians.

04 **Quantify the interval, not just the point.** Pinball loss and empirical coverage for every published range; conformal intervals where coverage must be guaranteed.

05 **Compare against the incumbent.** Every metric is reported beside today's interpolation or rule, with a confidence interval on the difference.

06 **Clinical review of errors.** The clinical lead reviews the worst 20 predictions, not just the averages. A systematic failure mode blocks adoption regardless of mean error.

07 **Prospective silent run.** The model runs beside the unit for an agreed period, recording what it would have said, with no influence on care, before anything is displayed.

TABLE 6 — METRICS BY PROBLEM

Problem	Primary	Secondary	Guardrail
P1 Length of stay	Median absolute error (days)	Concordance; calibration slope; Brier score for outcome	No band with error worse than today's method
P2 Acuity trajectory	Balanced accuracy	Confusion matrix by level; cost-weighted error	Under-calling rate below an agreed ceiling
P3 Demand	Mean absolute error (nurse-hours)	Pinball loss; interval coverage	Never below seasonal naive on any horizon
P4 Roster	Uncovered cots	Constraint violations; planner acceptance	Solve inside the interaction budget, or fall back

Reporting and governance

Reporting standards

Model development and validation reported to **TRIPOD+AI**; risk of bias assessed with **PROBAST**; any prospective evaluation follows **DECIDE-AI**. Each released model carries a one-page model card: intended use, data, performance by subgroup, known failure modes, and the date it must be reviewed.

Change control

Models are versioned artefacts with the data snapshot that produced them. A model change follows the same review, test and approval path as a code change, and the clinical lead signs the model card before release.

Monitoring in service

Input drift (case mix, missingness), output drift (distribution of predictions), and realised error once outcomes are known. Alert thresholds agreed per branch; breach triggers review, not silent retraining.

Human factors

Predictions are shown with their range and confidence, never as a single authoritative number. The interface requires an explicit human action for every staffing decision, and records that the recommendation was advisory.

Fairness and equity. Performance is reported by gestational-age band, birth weight, inborn versus outborn, and branch. A model that is materially worse for the smallest babies, or for one site, is not released even if its average is better.

08 How each role uses NeoStay

Six roles, each with the task they come to do and the path they take.

ROLE 1 · CHARGE NURSE / SHIFT COORDINATOR — THE PRIMARY USER

"How many nurses do I need tonight, and who should take which cot?"

When: at each shift handover, and whenever a cot changes acuity. **Time budget:** under five minutes for a full unit.

1. Sign in; the unit's census from the previous shift is the starting point.
2. For each cot, tick the most intensive finding per body system, the 1:1 box if the baby meets the footnoted criteria, and the highest level-of-care criterion met. The preview shows the ratio, care level and minimum nurse level before the cot is added.
3. Enter the nurses actually available by grade, and the shift pattern.
4. Read the verdict: nurses per shift, recommended skill mix, effective capacity against demand, and the 24-hour roster. Any cot without a qualified nurse is named in red, with the grade and number of extra nurses that would close the gap.
5. Export or record the decision; escalate with the figure if cover is short.

ROLE 2 · NEONATOLOGIST / CONSULTANT

"What should I tell these parents, and what will this baby need?"

Opens **Predict**, enters birth weight, gestational age, condition and respiratory support. Reads the expected stay with its realistic range, the chance of going home, transfer or death, the survival figure, and the confidence grade with the number of comparable babies behind it. Uses the per-phase nursing profile when planning cot moves and discharge preparation. Every figure carries the caveat that it describes groups, not this baby.

ROLE 3 · UNIT MANAGER / MATRON

"Is my establishment right, and what do I take to the staffing review?"

Runs the unit's typical census through the staffing model to compare recommended establishment with actual staffing; collects the record of shifts where cover was short and by which grade. Once the hospital supplies its own census history (D4), the same comparison runs over a real period. The output is evidence for an establishment case, not a rota.

ROLE 4 · PRACTICE EDUCATOR

"Who needs supervision, and where are my skill gaps?"

Reads the skill-mix panel's preceptor requirement and novice share across shifts, and the Acuity tool page as a teaching reference for the unit's own policy.

ROLE 5 · NEOSTAY ADMINISTRATOR (NURSING OR INFORMATICS)

"Who has access, and is the service healthy?"

Creates and removes accounts, resets passwords, checks the health endpoint and the service status. Account changes take effect immediately; removing an account signs that person out everywhere.

ROLE 6 · HOSPITAL IT / PLATFORM TEAM

"How do I run, secure, upgrade and back this up?"

Runs one container behind the hospital's HTTPS entry point, supplies five environment variables, backs up the accounts file, applies upgrades by deploying a new image tag, and monitors `/api/health`. Hospital single sign-on replaces NeoStay passwords where required.

Service expectations each role can rely on

Expectation	Target
Page ready to use after sign-in	Under 3 seconds on a ward computer
Census recalculated after a change	Immediate — all calculation is local to the browser
Availability during a shift	99.5% monthly, excluding planned windows
Support response for a clinical-hours issue	Same working day
Account created or removed	Within one working day of an approved request

09 Design and accessibility standards

The rules the interface is held to, and why they matter at 03:00.

Clarity under pressure

One primary action per screen. Numbers before adjectives. Plain verdicts in words ("fully covered", "short by 0.1 nurses") beside every chart, so a decision never depends on reading a graph correctly.

Traceability

Every derived number can be hovered or tapped to see what produced it — the findings that set a ratio, the sample size behind a forecast.

No silent failure

Missing data, low confidence, an expired session or a browser that cannot decrypt the published copy each produce a specific message and a next step.

Accessibility

Target **WCAG 2.2 AA**: contrast, focus visibility, full keyboard operation, labelled controls, text that scales to 200%, and no meaning carried by colour alone. Formal audit is a roadmap item.

Works on ward hardware

Responsive from 360 px upward; no dependency on hover; touch targets sized for gloved hands; fonts fall back cleanly if external fonts are blocked.

Consistency

One design system across landing page, sign-in and the application, so the product feels like one tool rather than three.

Usability engineering. For a clinical tool, usability is a safety property. The project follows the spirit of **IEC 62366-1**: named user groups, defined use scenarios, identification of use errors that could cause harm (for example a misread ratio), and testing with real users before wide release.

10 Engineering standards and quality gates

What must be true before any change reaches a ward.

TABLE 7 — STANDARDS IN FORCE

Area	Standard applied	Evidence
Application architecture	Twelve-factor service: configuration from the environment, stateless processes, logs to standard output, one build promoted across environments	Dockerfile, `docs/CLOUD.md`
Source control	Trunk-based with short-lived branches; every change reviewed; signed, descriptive commits; semantic versioning for releases	Repository history
Testing	Automated suite covering sign-in and sessions, published-site encryption, page gating, cloud configuration, and engine parity between the browser and Python implementations	105 tests currently passing
Continuous integration	Tests plus a real container build, started and smoke-tested end to end on every push	`.github/workflows/ci.yml`
Security	Hashed credentials, signed expiring sessions, rate limiting, strict Content-Security-Policy, non-root container, read-only root filesystem, secrets never in images or Git	`SECURITY.md`
Supply chain	Pinned dependencies, automated dependency updates, multi-architecture images with provenance, image digests pinned in production	Workflows, `requirements*.txt`
Documentation	User guide, deployment guide, cloud guide, architecture reference; every setting documented where it is defined	`docs/`
Clinical risk	Hazard log with severity, likelihood and mitigation, maintained with the clinical lead; aligned to the intent of **DCB0129/DCB0160** for health-IT risk management	Roadmap item, Section 15
Interoperability	Where integration is added, **HL7 FHIR** resources and **SNOMED CT** coding rather than bespoke formats	Roadmap item

Definition of done, for any change

- ☐ Tests written and passing, including a case that would have caught the defect.
- ☐ Browser and Python engines still agree, value for value.
- ☐ Security review for anything touching accounts, sessions or published data.
- ☐ Documentation updated in the same change, not afterwards.
- ☐ Clinical lead's approval for anything that alters a displayed clinical number or wording.
- ☐ Rollback path stated in the release note.

11 Installing across a multi-branch hospital group

The part most often underestimated: several units, one service, different local policies.

Choose a tenancy model first

TABLE 8 — THREE WAYS TO SERVE SEVERAL BRANCHES

Model	How it works	Best when	Trade-off
A · One deployment per branch	Each unit runs its own container and its own accounts; no shared data.	Branches are independent legal entities, or governance forbids any pooling.	No group-level view; upgrades repeated per site.
B · One deployment, branch-scoped configuration recommended	One service; each user belongs to a branch; policy, grades and ratios are per branch; data stays separated by branch.	A group with shared IT and a wish to compare units.	Requires the configuration work in Section 15, Phase 2.
C · Central service, federated data	Branches keep their own data; only aggregated statistics travel to the central service.	Cross-border groups, or where data residency differs by site.	Most complex; needed only if data cannot be centralised.

What must be configured per branch

Clinical policy

- Nurse grade ladder and the mapping to the four levels
- Ratios for each acuity column, and the local 1:1 criteria
- Levels of care in use, and the mapping to minimum nurse level
- Supervision rules: novice cap, preceptor requirement
- Shift pattern (two 12-hour, three 8-hour, local variants)

Operational

- Branch name, designated unit level, cot capacity
- Accounts and roles, by branch
- Host name and certificate
- Which dataset the branch's forecasts use — its own, or the group's pooled statistics
- Support contacts and escalation path

Installation checklist (per branch)

Before install

- ☐ Information-governance approval recorded
- ☐ Tenancy model agreed (A, B or C)
- ☐ Host name allocated; TLS certificate issued
- ☐ Nurse grade mapping signed off by the branch's nursing lead
- ☐ Acuity policy confirmed, or the group policy adopted explicitly
- ☐ Named administrator and deputy identified
- ☐ Account list agreed with line managers

Install and verify

- ☐ Container deployed; health check green
- ☐ HTTPS verified from a ward computer and a phone
- ☐ First administrator account created; password delivered securely
- ☐ End-to-end smoke test passed and recorded
- ☐ Branch configuration applied and reviewed
- ☐ Backup of the accounts file tested, including restore
- ☐ Monitoring and alerting connected

Go live

- ☐ Two training sessions delivered per shift pattern (day and night)
- ☐ Quick-reference card at each ward computer
- ☐ Two weeks of parallel running against the current manual method
- ☐ Daily check-in during week one; weekly thereafter
- ☐ Feedback log open, with named owner

Steady state

- ☐ Quarterly review of policy configuration with the nursing lead
- ☐ Access list reviewed every six months
- ☐ Upgrades applied within one month of release
- ☐ Incident and near-miss log reviewed with the clinical lead
- ☐ Annual revalidation of forecasts against that branch's outcomes

The three mistakes that derail multi-site rollouts. First, assuming one branch's ratios are everyone's — they are not, and the mapping must be signed per site. Second, creating accounts centrally without a local owner, so leavers keep access. Third, going live without parallel running, which removes the only opportunity to compare the recommendation with what the unit would have done anyway.

12 Scaling as usage grows

From one unit to a hospital group, without re-architecting.

NeoStay's calculations run in the browser, so user growth costs the server almost nothing: each additional user adds a sign-in check and a few static file requests. The scaling work is therefore about availability, data volume and the features that will add server-side state.

TABLE 9 — CAPACITY PLAN

Stage	Users	Shape	What changes
Pilot	Up to 50	One container, one replica, 0.5 vCPU, 512 MB	Nothing — today's deployment.
One hospital	50–300	2–3 replicas behind the hospital's load balancer; static files cached at the edge	Stateless mode with a shared session secret; health-based autoscaling.
Group of branches	300–2,000	Replicas per region or per branch; shared configuration service	Branch-scoped configuration and roles; a managed Postgres for saved censuses and audit; rate limiting moved to a shared store.
Group with analytics	2,000+	Separate read path for reporting; scheduled model retraining jobs	A warehouse for historical data; a job runner for forecasts; per-branch dashboards.

> **Known constraint to fix before large scale.** The limit on failed sign-ins is counted inside each server process, so several replicas allow proportionally more attempts. Moving that counter to a shared store is a prerequisite for an internet-facing deployment at group scale.

Performance budget

Operation	Budget	Why that number
Sign-in (server check)	< 500 ms	Perceived as instant at handover.
Census recalculation after a change	< 100 ms	Runs locally; must feel immediate while a nurse ticks findings.
Roster for 30 cots and 40 nurses	< 1 s heuristic; < 5 s solver	Beyond this, the planner stops trusting the tool and uses paper.
Forecast refresh (server job)	< 15 min nightly	Must finish well before the early shift.

What stays true as it grows

- **Stateless web tier.** Sessions are signed tokens, so any replica serves any user.
- **One image everywhere.** Branch differences are configuration, never a separate build.
- **Data stays small.** Statistics and policy files are kilobytes; only the optional historical warehouse grows, outside the request path.
- **Graceful degradation.** If a model service is down, the deterministic method still answers, labelled as such.

13 Information governance, safety and compliance

What must be in place before patient-level data is used, and where the device boundary sits.

TABLE 10 — GOVERNANCE REQUIREMENTS

Requirement	What it means here	Owner
Lawful basis and approvals	Service evaluation or research classification agreed; ethics approval where required; Caldicott Guardian or equivalent sign-off for any patient-level extract.	Clinical Lead with hospital IG
Data protection impact assessment	Completed before the first patient-level extract; records data flows, minimisation, retention and risks.	Hospital IG with Technical Lead
Data sharing agreement	Scope, purpose, pseudonymisation, security measures, retention and destruction, named recipients.	Hospital legal / IG
Minimisation and pseudonymisation	No names, addresses, NHS or MRN numbers. A study identifier per infant; dates reduced to the granularity actually needed.	Data provider
Security of transfer and storage	Encrypted transfer, access limited to named analysts, encryption at rest, no copies on personal devices.	Technical Lead
Retention and destruction	Agreed period; certified destruction at the end; model artefacts retain no raw records.	Both leads
Clinical safety	Hazard log, clinical safety case and sign-off before any clinical use, in the spirit of DCB0129/DCB0160.	Clinical Lead
Security assurance	Penetration test and dependency audit before wide release; annual repeat.	Technical Lead
Accessibility	WCAG 2.2 AA audit and remediation before group-wide rollout.	Technical Lead

The regulatory boundary. NeoStay today informs staffing and planning: it makes no diagnostic or treatment claim and is therefore not a medical device. Three changes would cross that line: predicting an individual baby's clinical outcome for clinical management, driving treatment decisions automatically, or marketing it for diagnosis or prognosis. If the project chooses to cross it, a quality management system (ISO 13485), software lifecycle process (IEC 62304), clinical evaluation and regulatory submission become mandatory, and the roadmap must be replanned accordingly.

14 Risks and how they are managed

Named owners, concrete mitigations.

TABLE 11 — RISK REGISTER (EXTRACT)

Risk	Impact	Mitigation
Data never arrives, or arrives incomplete	Models cannot be built; the product stays deterministic.	The companion requirements document makes the ask concrete and staged; a minimum viable extract is defined so work can start small.
Over-trust in a number	Staffing set by a forecast that was wrong for this unit.	Ranges and confidence always shown; wording states the figure is advisory; parallel running before go-live; audit of overridden recommendations.
One unit's practice baked into a model	Poor transfer to other branches.	Leave-one-branch-out validation; hierarchical models with partial pooling; per-branch calibration reports.
Policy drift	Ratios or grades change locally and the tool silently disagrees.	Configuration is versioned and dated; quarterly review with the nursing lead; the application shows which policy version it is using.
Access creep	Leavers retain accounts.	Six-monthly access review; single sign-on where available; immediate revocation on request.
Key-person dependency	Project stalls if one person is unavailable.	Everything in version control with written guides; second administrator per branch; documented deployment anyone competent can repeat.
Scope creep towards a medical device	Unplanned regulatory burden.	Device boundary stated (Section 13); any proposal that crosses it triggers a formal decision, not a quiet implementation.

15 Delivery plan

Three phases, each with an exit test.

PHASE 1 · FOUNDATIONS

Configure, pilot, measure

Branch-level configuration of grades, ratios and policy; accessibility and security audits; hazard log opened; pilot in one unit with parallel running; usage and task-time measured.

Exit test: one unit uses NeoStay for every shift for four weeks, with no safety incident and a majority of charge nurses preferring it to the manual method.

PHASE 2 · EVIDENCE

Data, models, validation

Patient-level extract under governance; P1 and P3 models built and compared with the deterministic baseline; silent prospective run; TRIPOD+AI report; model cards reviewed and signed.

Exit test: a model beats the incumbent on the primary metric and on calibration, in a temporal hold-out and in one external branch.

PHASE 3 · SCALE

Group rollout

Branch-scoped multi-tenancy; roster optimisation with constraints; saved censuses and audit; single sign-on; dashboards per branch; annual revalidation cycle established.

Exit test: a second and third branch onboarded by configuration only, each inside two weeks.

TABLE 12 — WHAT IS NEEDED FROM WHOM

From	What	When
Hospital management	Agreement in principle; named data owner; access to historical data under governance	Phase 1 start
Nursing leadership	Grade mapping, ratio policy, supervision rules, pilot unit and champions	Phase 1
Clinical lead	Policy sign-off, error review, model card approval, hazard log	Throughout
Informatics / IT	Data extract, hosting, HTTPS, single sign-on, monitoring	Phases 1–2
Technical lead	Configuration, models, validation reports, deployment and documentation	Throughout

16 Appendices

Reference material for reviewers.

A · Glossary of methods named in this document

Term	Plain meaning
Survival model (AFT, Cox)	A model for "how long until something happens" that copes with cases still ongoing when the data was extracted.
Competing risks	More than one way the stay can end. Treating death as if the baby were simply still inpatient would overstate length of stay.
Multi-state model	A model of movement between care levels, rather than a single endpoint.
Calibration	Whether predicted numbers match observed reality on average: if the tool says 100 days, do such babies stay about 100 days?
Concordance (C-index)	Whether the model ranks babies correctly: does the one it says will stay longer actually stay longer?
Pinball loss	A score for a predicted range rather than a single number.
Conformal prediction	A method that produces ranges with a guaranteed hit rate, whatever model sits underneath.
Hierarchical (partial pooling)	Letting a small unit borrow strength from the group while keeping its own signal.
Constraint programming	Describing the rules a roster must obey and letting a solver find an assignment that satisfies them.
Drift	The world changing after a model was built — different case mix, different practice — so its accuracy decays.

B · Companion documents

Document	Purpose	Audience
Executive brief	Four pages: the ask, the plan, the decisions needed.	Hospital executive, service managers
Data & feature requirements	Exactly which data is requested, in what format and volume, with governance steps and sign-off.	Hospital management, informatics, via the Clinical Lead
Architecture & methods reference	How the current system works, algorithm by algorithm, with worked examples.	Technical reviewers, new engineers
Deployment and cloud guides	Step-by-step installation on a hospital server or any cloud.	Hospital IT

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